SteadyStud – Magnetic Stud Finer (Aluminum) – RFQ Spec (Qty 100)

Part name: Magnetic Stud Finer v1
CAD: `Magnetic Stud Finer v1.step` (STEP AP242, units = **inch**)
Qty: 100 pcs (please also quote price breaks if convenient: 50 / 100 / 250)
Material: Aluminum (vendor-recommended; see questions below)
Magnet: Do not include magnet in this RFQ. Magnet(s) will be sourced separately; however, machining features must be compatible with the intended magnet placement.

1) Overview

Small aluminum part intended to interface with a separately-sourced magnet. Includes a **laser-etched marking** on one face.

We are requesting quotes for **two manufacturing routes**:

Option A — CNC from billet (per STEP)

- Manufacture per provided STEP model.
- Deburr and finish suitable for handling (no sharp edges).

Option B — Lower-cost plate/sheet + profiling + minimal secondary ops (if feasible)

- Goal: reduce cost versus full 3D CNC by using plate stock and 2D profiling (laser / waterjet) plus minimal machining.
- Must maintain the **same outer profile and critical interface dimensions** as Option A.
- Vendor to advise feasibility and any required design tweaks.

2) Critical dimensions / interfaces (callouts)

Use STEP as the governing geometry. The following are called out as **critical** and should be inspected:

1. **Overall length (L)** – along the long axis of the part (per STEP bounding length).
2. **Overall width (W)** – across the main body (per STEP bounding width).
3. **Top notch / key feature** – the top notch geometry and its location relative to ends/edges (per STEP).

   Note: We do not have a 2D drawing yet; please quote based on the STEP and confirm what tolerances you can reliably hold. If you require a 2D drawing with GD&T, list exactly what you need and we will provide.

3) Tolerances (initial request)

- **Default (non-critical):** ±0.010 in (or vendor standard)
- **Critical features (L/W + top notch):** target **±0.005 in** if achievable cost-effectively
- Flatness/parallelism: vendor standard unless you recommend otherwise for the function.

If these tolerances are unrealistic for Option B, propose an alternate tolerance set and explain impact on function and cost.

4) Thickness / stock

- STEP file units are inches. The model Z extent appears small (plate-like). Please advise:
  - Recommended **nominal thickness** for strength and manufacturability
  - Best stock form (plate, bar, etc.)

5) Edge condition & deburr

- Break all sharp edges.
- No burrs or sharp corners.

6) Marking (laser etch)

- **Laser etch required** (simple text or logo).
- Please quote as an add-on line item and specify:
  - Minimum feature size / contrast
  - Preferred file formats (DXF/SVG/AI/etc.)
  - Any setup fees

7) Finish

- Please quote:
  - **As-machined** (deburred)
  - Optional: **Type II anodize** (clear or black) as an alternate line item

8) Deliverables

- Quote for Option A and Option B (if feasible), each including:
  - Unit price, setup/NRE if any
  - Lead time (prototype/first article + full 100 pcs)
  - Inspection approach for critical dims
- Confirm you can accept STEP AP242.

9) Vendor questions (please answer in your quote)

1. What alloy do you recommend (e.g., 6061-T6 vs 7075 vs other) and why?
2. For **Option B**, is a plate+profile approach feasible while maintaining critical geometry? What secondary ops would be required?
3. What tolerances can you hold on the **overall L/W** and **top notch** features for each option?
4. What thickness/stock would you choose? Any warping concerns (especially for profiling routes)?
5. Preferred finishing (as-machined vs anodize). Any impact on dimensions?
6. Laser etch: equipment type, typical cost adders, and best location on the part.
7. Do you recommend adding datum structure / a 2D drawing? If yes, what callouts are required?
8. What is your typical yield/scrap expectation for Qty 100?
9. Shipping terms/location (FOB), packaging recommendations to avoid marring.

Attachments (provided): - `Magnetic Stud Finer v1.step` - `IMG_4372.MOV` (reference video) - `ScreenRecording_03-11-2025 21-00-17_1.MP4` (reference video)